Lower and Upper Decomposition of Matrix Or Matrix Factorization

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Lower and Upper Decomposition of Matrix

This method tells us, the decomposition of only square matrix. The decomposition stated that any square matrix is broken as product of two matrices.

$$A = LU$$

Where A is any square matrix, L is lower triangular matrix, U is upper triangular matrix.

The matrix factorization consists three method.

(1) Crout's Method

(2) Doolittle's Method

(3) Choleskey's Method

How to work the LU Decomposition

Let the system

$$AX = b \to (i)$$

We decompose A by $A = LU$ put in eq. (i)

$$LUX = b \to (ii)$$

Suppose $UX = Y \to (iii)$

Put (iii) in (ii)

$$LY = b \to (iv)$$

From (iii) we get values of Y and put in (iii), to get the unknown value of X.

(1) Crout's Method

This method decomposes a square matrix into product of two matrices. That composition $A = LU$, where A is square matrix, L is lower triangular matrix, U is the upper triangular matrix.

Mathematically,

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} 1 & u_{12} & u_{13} \\ 0 & 1 & u_{23} \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{bmatrix}$$

Note: In Crout's method, upper diagonal matrix has ones diagonally.

(2) Doolittle's Method

This method decomposes a square matrix into the product of two matrices, $A = LU$.

Mathematically

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

Note: In Doolittle's method, lower diagonal matrix has ones diagonally.

(3) Choleskey's Method

This method decomposes a square matrix into the product of two matrices, $A = LU$. If main diagonal in L and U are not taken unity is called Choleskey's Method.

Mathematically,

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} l_{11}u_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + l_{22}u_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + l_{33}u_{33} \end{bmatrix}$$

Q#1: Solve the system of linear equations using Crout's Method.

$$2x_1 - 2x_2 + 4x_3 = 0$$
$$x_1 - 3x_2 + x_3 = -5$$
$$3x_1 + 7x_2 + 5x_3 = 7$$

Sol:

$$\begin{bmatrix} 2 & -2 & 4 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 7 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & -2 & 4 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix}$$
$$A = LU$$
$$\begin{bmatrix} 2 & -2 & 4 \\ 1 & -3 & 1 \\ 3 & 7 & 5 \end{bmatrix} = \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{bmatrix}$$

$$l_{11} = 2$$
$$l_{21} = 1$$
$$l_{31} = 3$$
$$l_{11}u_{12} = -2$$
$$u_{12} = -1$$
$$l_{11}u_{13} = 4$$
$$u_{13} = 2$$
$$l_{21}u_{12} + l_{22} = -3$$
$$(1)(-1) + l_{22} = -3$$
$$l_{22} = -2$$

$$l_{21}u_{13} + \ l_{22}u_{23} = 1$$
$$(1)(2) + (-2)u_{23} = 1$$
$$u_{23} = \frac{1}{2}$$
$$l_{31}u_{12} + \ l_{32} = 7$$
$$(3)(-1) + \ l_{32} = 7$$
$$l_{32} = 10$$
$$l_{31}u_{13} + \ l_{32}u_{23} + l_{33} = 5$$
$$(3)(2) + (10)\left(\frac{1}{2}\right) + l_{33} = 5$$
$$l_{33} = -6$$

Here $L = \begin{bmatrix} 2 & 0 & 0 \\ 1 & -2 & 0 \\ 3 & 10 & -6 \end{bmatrix}$ and $U = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix}$

$$AX = LUX = b$$

So $UX = Y$ then $LY = b$

$$LY = b$$

$$\begin{bmatrix} 2 & 0 & 0 \\ 1 & -2 & 0 \\ 3 & 10 & -6 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -5 \\ 7 \end{bmatrix}$$

$$2y_1 = 0 \rightarrow (i)$$

$$y_1 - 2y_2 = -5 \rightarrow (ii)$$

$$3y_1 + 10y_2 - 6y_3 = 7 \rightarrow (iii)$$

From (i) $y_1 = 0$ put in (ii)

$$0 - 2y_2 = -5 \Longrightarrow y_2 = \frac{5}{2}$$

Put $y_1 = 0$ and $y_2 = \frac{5}{2}$ in (iii)

$$3(0) + 10\left(\frac{5}{2}\right) - 6y_3 = 7$$

$$-6y_3 = -18$$

$$y_3 = 3$$

Now $UX = Y$

- $$\begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & \frac{1}{2} \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{5}{2} \\ 3 \end{bmatrix}$$

$$x_1 - x_2 + 2x_3 = 0 \rightarrow (iv)$$

$$x_2 + \frac{1}{2}x_3 = \frac{5}{2} \rightarrow (v)$$

$$x_3 = 3 \rightarrow (vi)$$

Put $x_3 = 3$ in (v)

$$x_2 + \frac{1}{2}(3) = \frac{5}{2}$$

$$x_2 = \frac{5}{2} - \frac{3}{2} = \frac{2}{2} = 1$$

Put $x_3 = 3$ and $x_2 = 1$ in (iv)

$$x_1 - 1 + 2(3) = 0$$

$$x_1 = -5$$

The solution set is $X = (-5, 1, 3)$.

Q#2: Solve the system of linear equations using Crout's Method.

$$4x_1 + 3x_2 - 5x_3 = 2$$
$$-4x_1 - 5x_2 + 7x_3 = -4$$
$$8x_1 + 6x_2 - 8x_3 = 6$$

Sol:

$$\begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$$

$$A = \begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix}$$

$$A = LU$$

$$\begin{bmatrix} 4 & 3 & -5 \\ -4 & -5 & 7 \\ 8 & 6 & -8 \end{bmatrix} = \begin{bmatrix} l_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21} & l_{21}u_{12} + l_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31} & l_{31}u_{12} + l_{32} & l_{31}u_{13} + l_{32}u_{23} + l_{33} \end{bmatrix}$$

$$l_{11} = 4$$
$$l_{21} = -4$$
$$l_{31} = 8$$
$$l_{11}u_{12} = 3$$
$$u_{12} = \frac{3}{4}$$
$$l_{11}u_{13} = -5$$
$$u_{13} = -\frac{5}{4}$$
$$l_{21}u_{12} + l_{22} = -5$$
$$(-4)\left(\frac{3}{4}\right) + l_{22} = -5$$
$$l_{22} = -2$$

$$l_{21}u_{13} + \ l_{22}u_{23} = 7$$

$$(-4)\left(-\frac{5}{4}\right) + (-2)u_{23} = 7$$

$$u_{23} = -1$$

$$l_{31}u_{12} + \ l_{32} = 6$$

$$(8)\left(\frac{3}{4}\right) + \ l_{32} = 6$$

$$l_{32} = 0$$

$$l_{31}u_{13} + \ l_{32}u_{23} + l_{33} = -8$$

$$(8)\left(-\frac{5}{4}\right) + (0)(-1) + l_{33} = -8$$

$$l_{33} = 2$$

- Here $L = \begin{bmatrix} 4 & 0 & 0 \\ -4 & -2 & 0 \\ 8 & 0 & 2 \end{bmatrix}$ and $U = \begin{bmatrix} 1 & \frac{[illegible]}{4} & -\frac{[illegible]}{4} \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

$$AX = b$$
$$LUX = b$$

So $UX = Y$ then $LY = b$

$$LY = b$$
$$\begin{bmatrix} 4 & 0 & 0 \\ -4 & -2 & 0 \\ 8 & 0 & 2 \end{bmatrix}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ 6 \end{bmatrix}$$
$$4y_1 = 2 \rightarrow (i)$$
$$-4y_1 - 2y_2 = -4 \rightarrow (ii)$$
$$8y_1 + 2y_3 = 6 \rightarrow (iii)$$

From (i) $y_1 = \frac{1}{2}$ put in (ii)

$$-4\left(\frac{1}{2}\right) - 2y_2 = -4 \Longrightarrow y_2 = 1$$

Put $y_1 = \frac{1}{2}$ and $y_2 = 1$ in (iii)

$$8\left(\frac{1}{2}\right) + 2y_3 = 6$$
$$2y_3 = 2$$
$$y_3 = 1$$
$$Y = \begin{bmatrix} \frac{1}{2} \\ 1 \\ 1 \end{bmatrix}$$

Now $UX = Y$

- $$\begin{bmatrix} 1 & \frac{3}{4} & -\frac{5}{4} \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ 1 \\ 1 \end{bmatrix}$$

$$x_1 + \frac{3}{4}x_2 - \frac{5}{4}x_3 = \frac{1}{2} \rightarrow (iv)$$
$$x_2 - x_3 = 1 \rightarrow (v)$$
$$x_3 = 1 \rightarrow (vi)$$

Put $x_3 = 1$ in (v)

$$x_2 - (1) = 1$$
$$x_2 = 2$$

Put $x_3 = 1$ and $x_2 = 2$ in (iv)

$$x_1 + \frac{3}{4}(2) - \frac{5}{4}(1) = \frac{1}{2}$$
$$x_1 = \frac{1}{4}$$

The solution set is $X = \left(\frac{1}{4}, 2, 1\right)$.

Q#3: Solve the system of linear equations using Doolittle's Method.

$$3x_1 - 7x_2 - 2x_3 = -7$$
$$-3x_1 + 5x_2 + x_3 = 5$$
$$6x_1 - 4x_2 = 2$$

Sol:

$$\begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -7 \\ 5 \\ 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix}$$

$$A = LU$$

$$\begin{bmatrix} 3 & -7 & -2 \\ -3 & 5 & 1 \\ 6 & -4 & 0 \end{bmatrix} = \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

$$u_{11} = 3$$
$$u_{12} = -7$$
$$u_{13} = -2$$
$$l_{21}u_{11} = -3$$
$$l_{21} = \frac{-3}{3} = -1$$
$$l_{21}u_{12} + u_{22} = 5$$
$$(-1)(-7) + u_{22} = 5$$
$$u_{22} = -2$$
$$l_{21}u_{13} + u_{23} = 1$$
$$(-1)(-2) + u_{23} = 1$$
$$u_{23} = -1$$
$$l_{31}u_{12} + l_{32}u_{22} = -4$$
$$(2)(-7) + l_{32}(-2) = -4$$
$$l_{32} = -5$$
$$l_{31}u_{13} + l_{32}u_{23} + u_{33} = 0$$
$$(2)(-2) + (-5)(-1) + u_{33} = 0$$
$$u_{33} = -1$$

- Here $L = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{bmatrix}$ and $U = \begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix}$

$$AX = b$$
$$LUX = b$$

So $UX = Y$ then $LY = b$

$$LY = b$$
$$\begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 2 & -5 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} -7 \\ 5 \\ 2 \end{bmatrix}$$
$$y_1 = -7 \rightarrow (i)$$
$$-y_1 + y_2 = 5 \rightarrow (ii)$$
$$2y_1 - 5y_2 + y_3 = 2 \rightarrow (iii)$$

From (i) $y_1 = -7$ put in (ii)

$$-(-7) + y_2 = 5 \Longrightarrow y_2 = -2$$

Put $y_1 = -7$ and $y_2 = -2$ in (iii)

$$2(-7) - 5(-2) + y_3 = 2$$
$$y_3 = 6$$
$$Y = \begin{bmatrix} -7 \\ -2 \\ 6 \end{bmatrix}$$

Now $UX = Y$

- $$\begin{bmatrix} 3 & -7 & -2 \\ 0 & -2 & -1 \\ 0 & 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -7 \\ -2 \\ 6 \end{bmatrix}$$

$$3x_1 - 7x_2 - 2x_3 = -7 \rightarrow (iv)$$
$$-2x_2 - x_3 = -2 \rightarrow (v)$$
$$-x_3 = 6 \rightarrow (vi)$$

Put $x_3 = -6$ in (v)

$$-2x_2 - \ (-6) = -2$$
$$-2x_2 = -8$$
$$x_2 = 4$$

Put $x_3 = -6$ and $x_2 = 4$ in (iv)

$$3x_1 - 7(4) - 2(-6) = -7$$
$$3x_1 - 28 + 12 = -7$$
$$3x_1 = 9$$
$$x_1 = 3$$

The solution set is $X = (3, 4, -6)$.

Q#4: Solve the system of linear equations using Choleskey's Method.

- $$2x_1 - x_2 + 2x_3 = 1$$
$$-6x_1 - 2x_3 = 0$$
$$8x_1 - x_2 + 5x_3 = 4$$

Sol:

$$\begin{bmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{bmatrix}$$

$$A = LU$$

$$\begin{bmatrix} 2 & -1 & 2 \\ -6 & 0 & -2 \\ 8 & -1 & 5 \end{bmatrix} = \begin{bmatrix} l_{11}u_{11} & l_{11}u_{12} & l_{11}u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + l_{22}u_{22} & l_{21}u_{13} + l_{22}u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + l_{33}u_{33} \end{bmatrix}$$

$$l_{11}u_{11} = 2$$
$$l_{11}u_{12} = -1$$
$$\frac{l_{11}u_{11}}{l_{11}u_{12}} = \frac{2}{-1}$$
$$\frac{u_{11}}{2} = \frac{u_{12}}{-1}$$
$$u_{11} = 2 \; and \; u_{12} = -1$$
$$l_{11}(2) = 2 \Rightarrow l_{11} = 1$$
$$l_{11}u_{13} = 2$$
$$u_{13} = 2$$
$$l_{21}u_{11} = -6$$
$$l_{21} = -3$$
$$l_{31}u_{11} = 8$$
$$l_{31} = 4$$
$$l_{21}u_{12} + l_{22}u_{22} = 0$$
$$(-3)(-1) + l_{22}u_{22} = 0$$
$$l_{22}u_{22} = -3$$

$$l_{21}u_{13} + l_{22}u_{23} = -2$$
$$(-3)(2) + l_{22}u_{23} = -2$$
$$l_{22}u_{23} = 4$$
$$\frac{l_{22}u_{22}}{l_{22}u_{23}} = \frac{-3}{4}$$
$$\frac{u_{22}}{u_{23}} = \frac{-3}{4}$$
$$\frac{u_{22}}{-3} = \frac{u_{23}}{4}$$
$$u_{22} = -3 \; and \; u_{23} = 4$$
$$l_{31}u_{12} + l_{32}u_{22} = -1$$
$$(4)(-1) + l_{32}(-3) = -1$$
$$l_{32} = 1$$
$$l_{31}u_{13} + l_{32}u_{23} + l_{33}u_{33} = 5$$
$$(4)(2) + (-1)(4) + l_{33}u_{33} = 5$$
$$l_{33}u_{33} = 1$$
$$\frac{l_{33}}{1} = \frac{u_{33}}{1}$$
$$l_{33} = 1 \; and \; u_{33} = 1$$

- Here $L = \begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -1 & 1 \end{bmatrix}$ and $U = \begin{bmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

$$AX = b$$
$$LUX = b$$

So $UX = Y$ then $LY = b$

$$LY = b$$
$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 4 & -1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}$$
$$y_1 = 1 \rightarrow (i)$$
$$-3y_1 + y_2 = 0 \rightarrow (ii)$$
$$4y_1 - y_2 + y_3 = 4 \rightarrow (iii)$$

From (i) $y_1 = 1$ put in (ii)

$$-3(1) + y_2 = 0 \Longrightarrow y_2 = 3$$

Put $y_1 = 1$ and $y_2 = 3$ in (iii)

$$4(1) - (3) + y_3 = 4$$
$$y_3 = 3$$
$$Y = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$$

Now $UX = Y$

- $$\begin{bmatrix} 2 & -1 & 2 \\ 0 & -3 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 3 \end{bmatrix}$$

$$2x_1 - x_2 + 2x_3 = 1 \rightarrow (iv)$$
$$-3x_2 + 4x_3 = 3 \rightarrow (v)$$
$$x_3 = 3 \rightarrow (vi)$$

Put $x_3 = 3$ in (v)

$$-3x_2 + 4(3) = 3$$
$$-3x_2 = -9$$
$$x_2 = 3$$

Put $x_3 = 3$ and $x_2 = 3$ in (iv)

$$2x_1 - (3) + 2(3) = 1$$
$$2x_1 - 3 + 6 = 1$$
$$2x_1 = -2$$
$$x_1 = -1$$

The solution set is $X = (-1, 3, 3)$.

THANK YOU

FOR YOUR ATTENTION

ANY QUESTIONS?